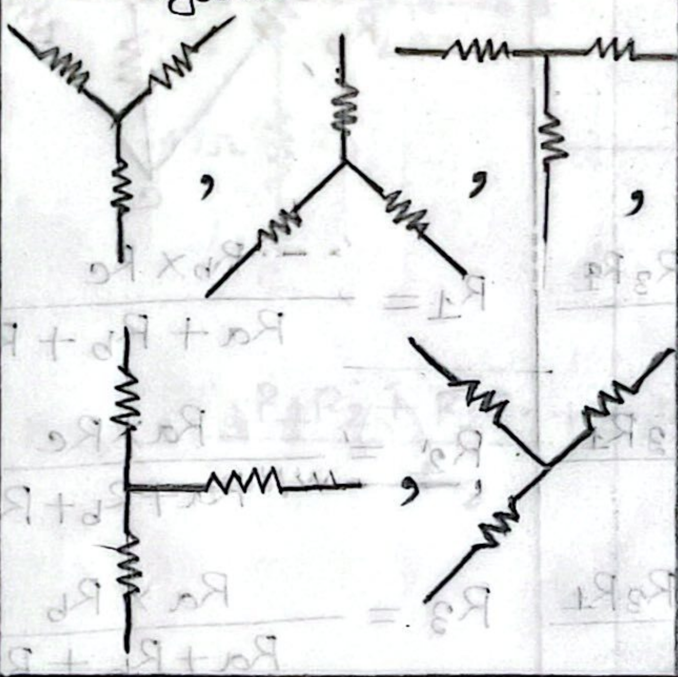
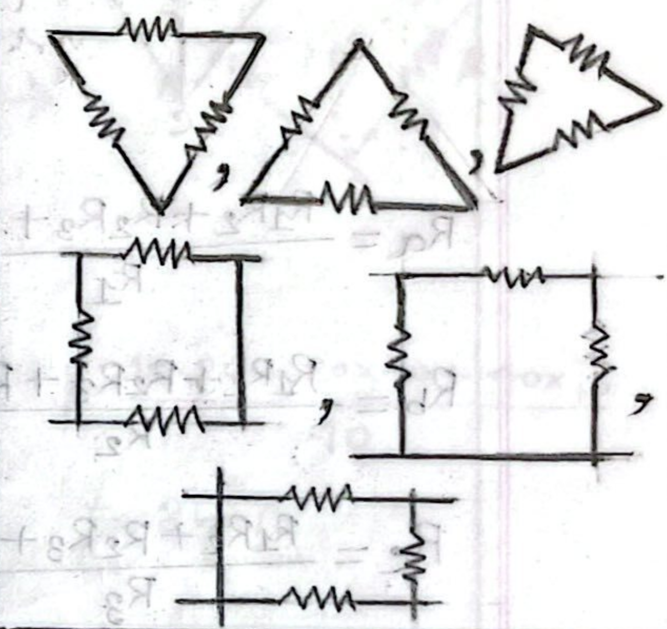
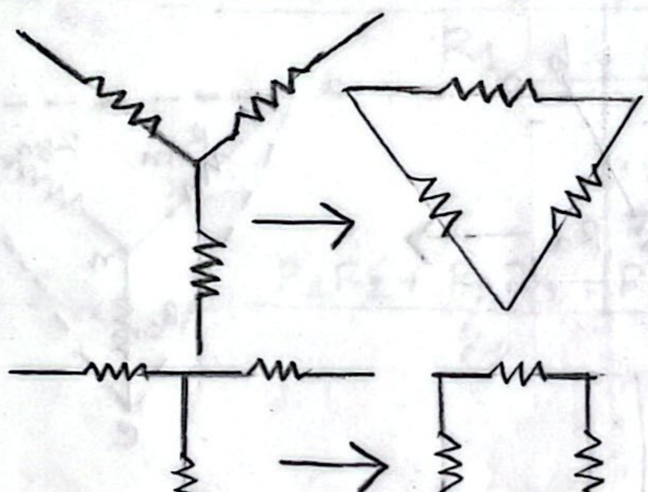
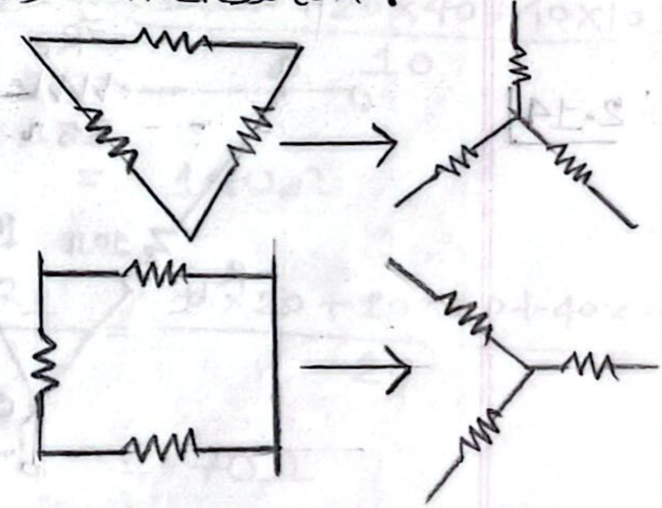
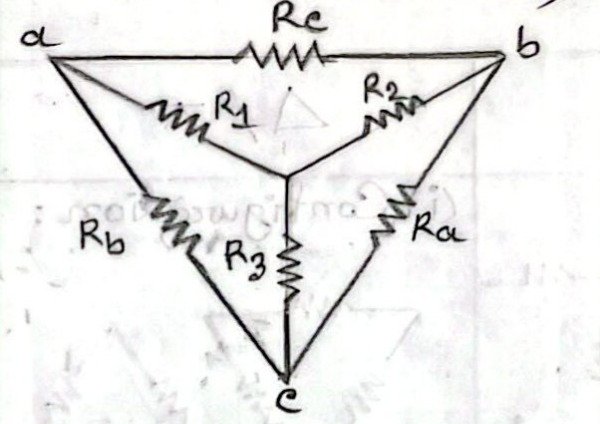
23.2.26

- **WYE-Delta Transformation -**

Y/T	Δ/π
(i) Configuration : 	(i) Configuration : 
(ii) Conversion : 	(ii) Conversiton : 
(iii) Looks like the letter Y / T.	(iii) Looks like a triangle (Δ) / π.
(iv) Three resistors connected to a common point.	(iv) Three resistors connected in a closed loop.

23.2.26

(v) Formula : $(Y \rightarrow \Delta)$

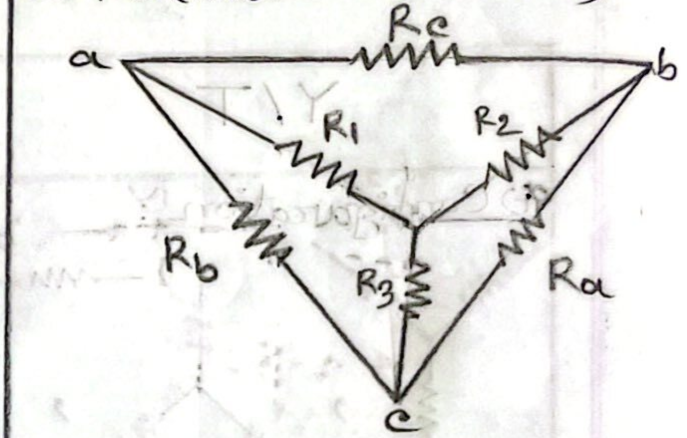


$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}$$

$$R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2}$$

$$R_c = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3}$$

(v) Formula : $(\Delta \rightarrow Y)$

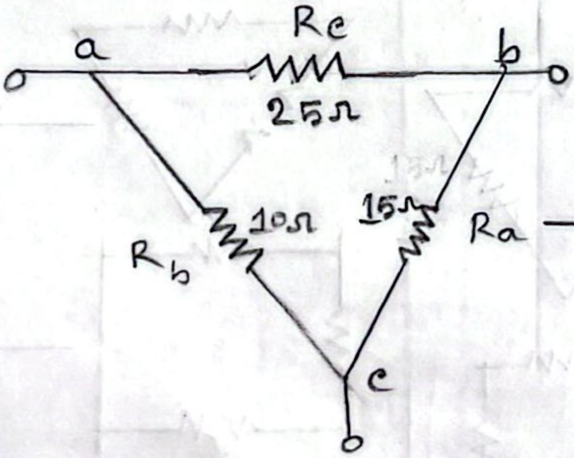


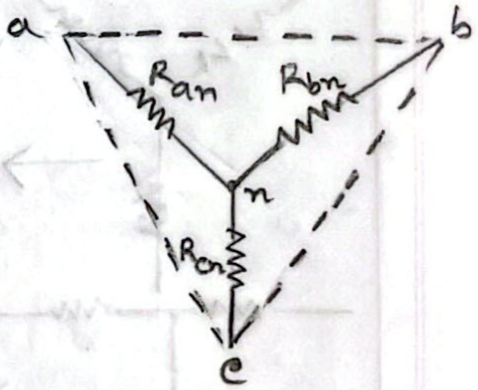
$$R_1 = \frac{R_b \times R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a \times R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a \times R_b}{R_a + R_b + R_c}$$

2-14





$$\therefore R_{an} = \frac{R_b \times R_c}{R_a + R_b + R_c} = \frac{10 \times 25}{10 + 15 + 25} = 5\Omega$$

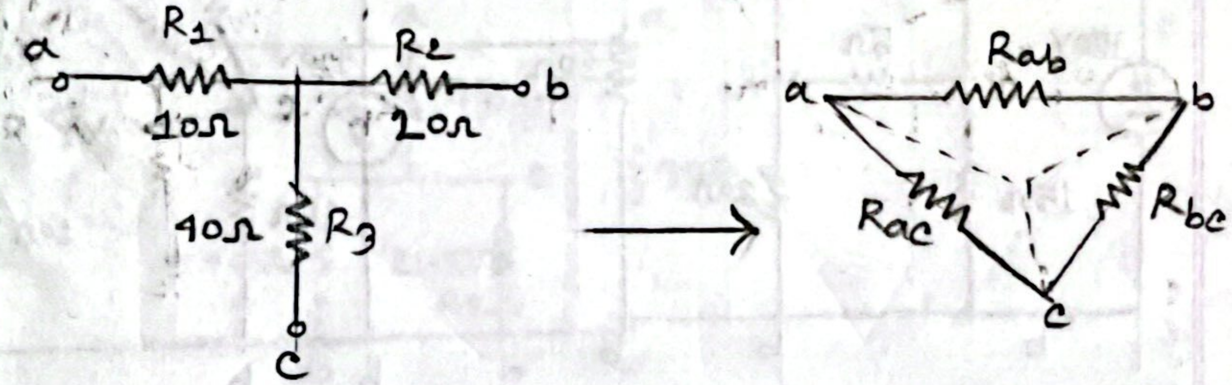
$$\therefore R_{bn} = \frac{R_a \times R_c}{R_a + R_b + R_c} = \frac{15 \times 25}{10 + 15 + 25} = 7.5\Omega$$

23.2.26

$$\therefore R_{cn} = \frac{R_b \times R_a}{R_a + R_b + R_c} = \frac{10 \times 15}{10+15+25} = 3\Omega$$

(Ans.)

P-2.14



$$\therefore R_{ab} = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3} = \frac{10\times20 + 20\times40 + 40\times10}{40}$$

$$= 35\Omega$$

$$\therefore R_{bc} = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1} = \frac{10\times20 + 20\times40 + 40\times10}{10}$$

$$= 140\Omega$$

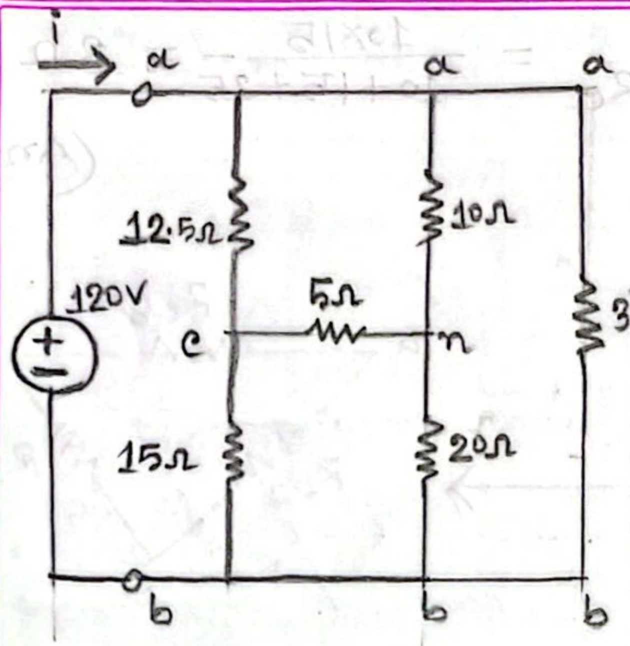
$$\therefore R_{ac} = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2} = \frac{10\times20 + 20\times40 + 40\times10}{20}$$

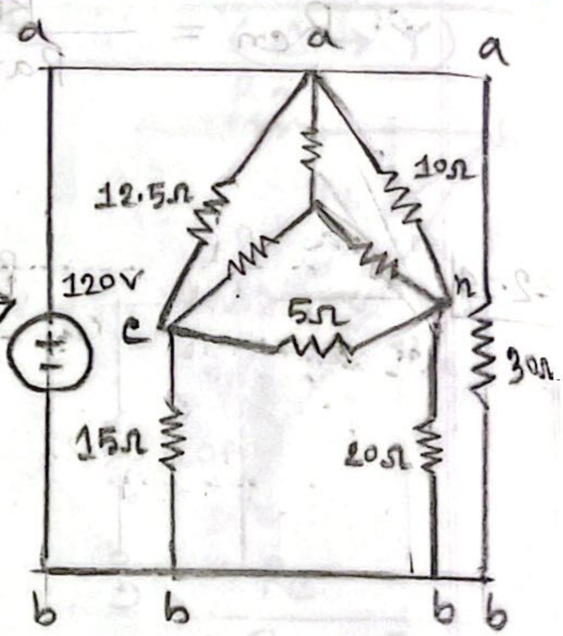
$$= 70\Omega$$

(Ans.)

Mid***

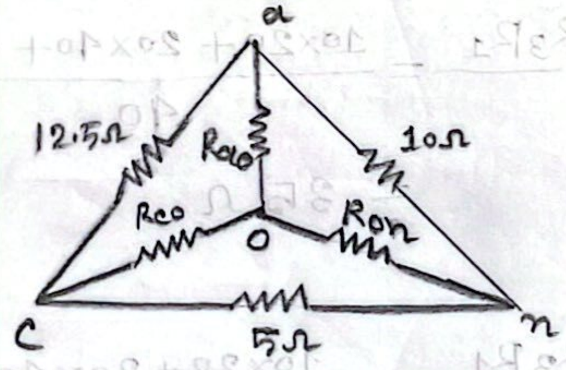
2.15





Find R_{ab} and i.

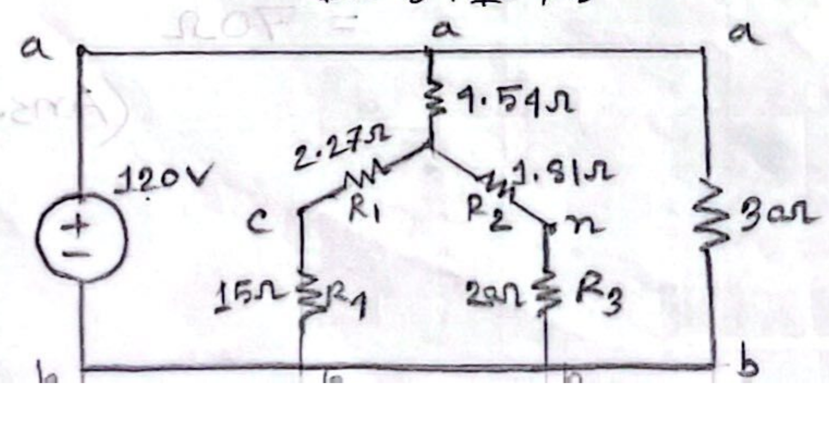
$(\Delta/\pi \rightarrow Y)$



$$\therefore R_{ao} = \frac{12.5 \times 10}{12.5 + 10 + 5} = 4.54\Omega$$

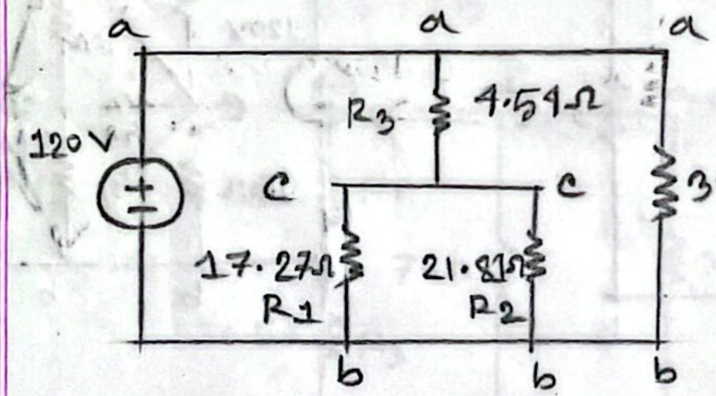
$$\therefore R_{co} = \frac{12.5 \times 5}{12.5 + 10 + 5} = 2.27\Omega$$

$$\therefore R_{on} = \frac{10 \times 5}{12.5 + 10 + 5} = 1.81\Omega$$



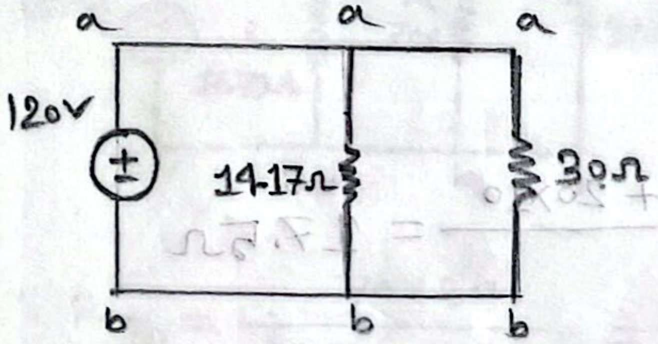
$\therefore R_1 = 2.27 + 15 = 17.27\,\Omega$

$\therefore R_2 = 1.81 + 20 = 21.81\,\Omega$



$$\therefore R_1 = \frac{17.27 \times 21.81}{17.27 + 21.81} = 9.63\,\Omega$$

$\therefore R_3 = 4.54 + 9.63 = 14.17\,\Omega$

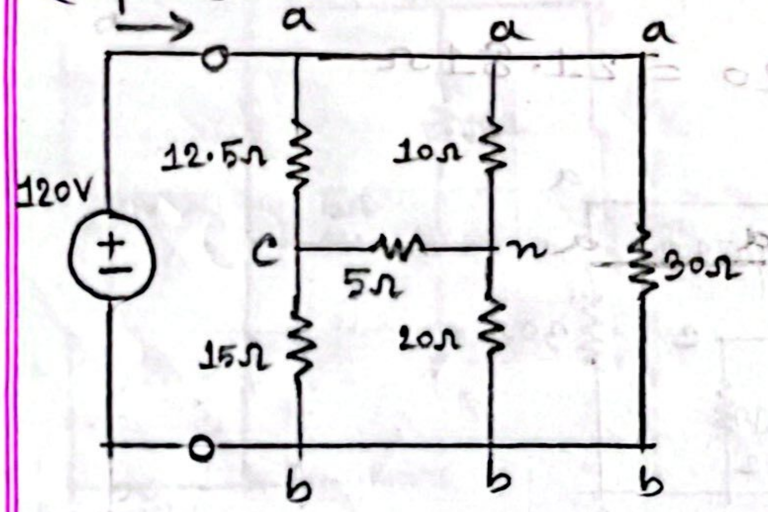


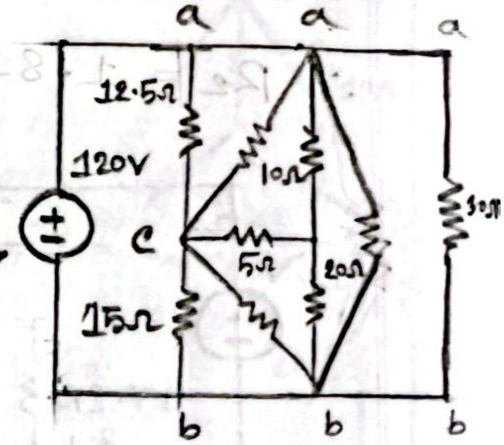
$$\therefore R_{ab} = \frac{30 \times 14.17}{30 + 14.17} = 9.62\,\Omega$$

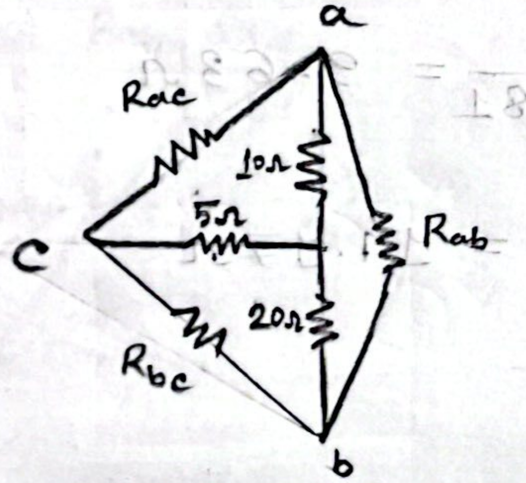
$$\therefore i = \frac{V}{R_{ab}} = \frac{120}{9.62} = 12.47\,A$$

(Ans.)

(Y/T → Δ)



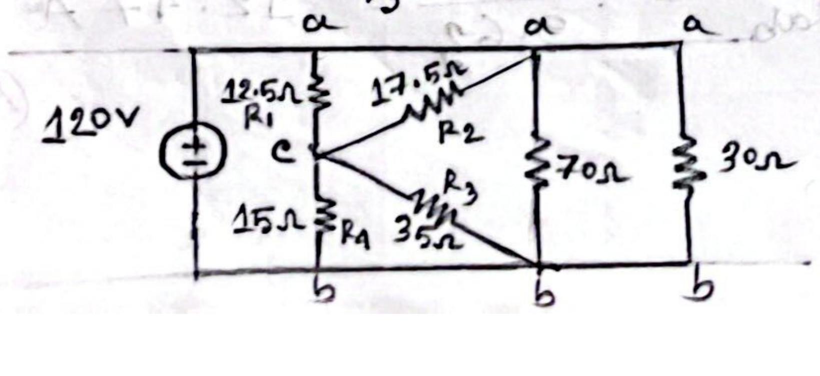




$$\therefore R_{ac} = \frac{10\times5+5\times20+20\times10}{20} = 17.5\Omega$$

$$\therefore R_{bc} = \frac{10\times5+5\times20+20\times10}{10} = 35\Omega$$

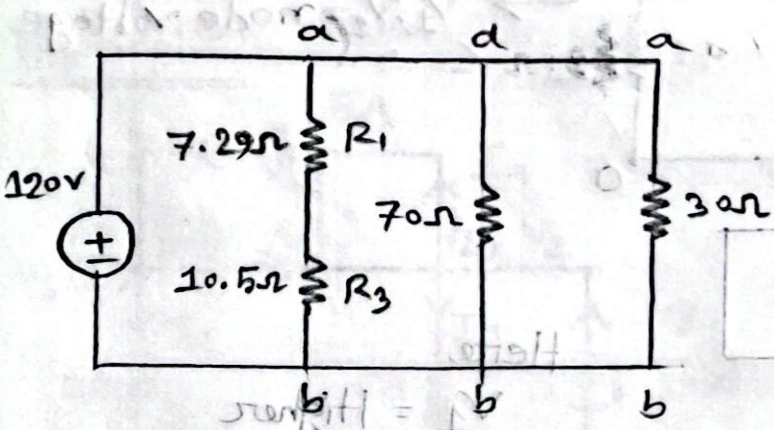
$$\therefore R_{ab} = \frac{10\times5+20\times5+20\times10}{5} = 70\Omega$$



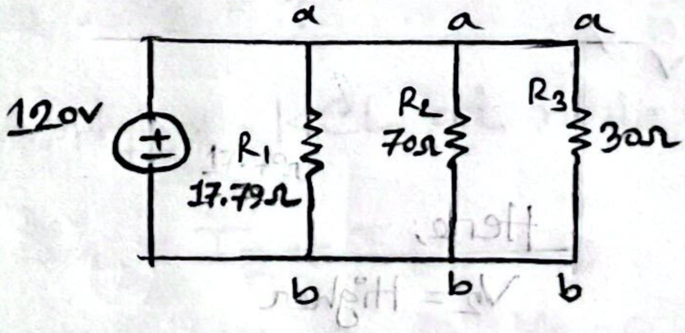
23.2.26

$$R_1 = \frac{12.5 \times 17.5}{12.5 + 17.5} = 7.29\Omega$$

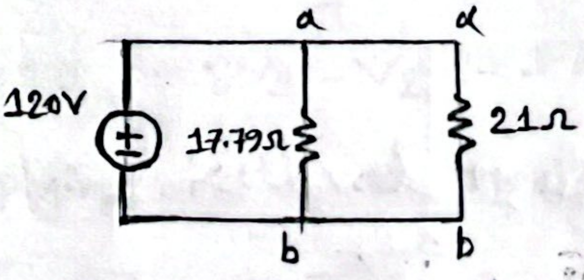
$$R_3 = \frac{35 \times 15}{35 + 15} = 10.5\Omega$$



$$\therefore R_1 = 7.29 + 10.5 = 17.79\Omega$$



$$\therefore R_{20} = \frac{70 \times 30}{70 + 30} = 21\Omega$$

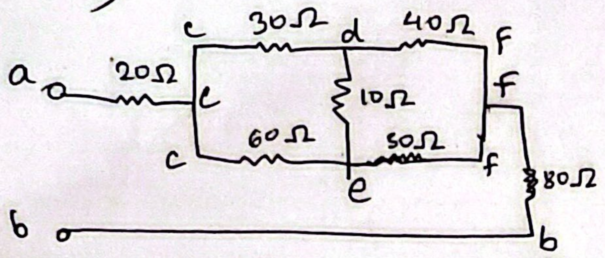


$$\therefore R_{ab} = \frac{21 \times 17.79}{17.79 + 21} = 9.62\Omega$$

$$\therefore i = \frac{V}{R_{ab}} = \frac{120}{9.62} = 12.47\ A$$ (Ans:)

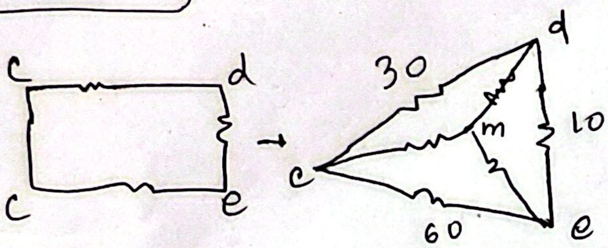
2.53

a)



Q: Obtain equivalent resistance Rab

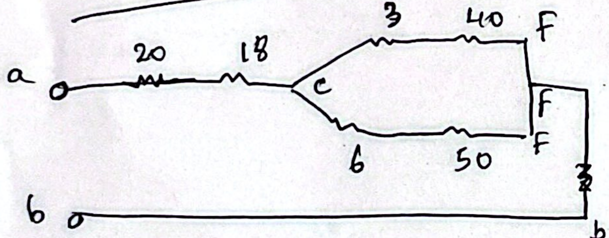
$\boxed{\Delta \rightarrow Y}$

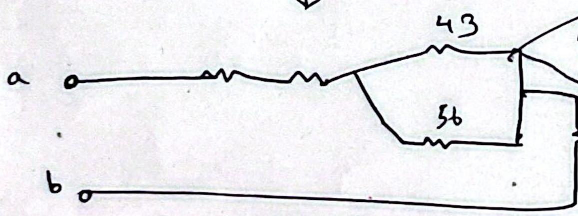


$$R_{cm} = \frac{30 \times 60}{30+60+10} = 18\,\Omega$$

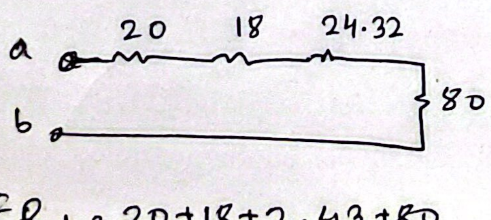
$$R_{dm} = \frac{30 \times 10}{100} = 3\,\Omega$$

$$R_{em} = \frac{10 \times 60}{100} = 6\,\Omega$$



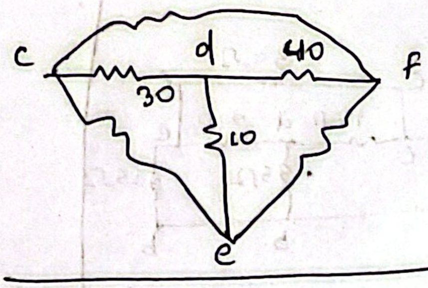


$$\frac{43 \times 56}{43+56}$$



$$ER_{ab} = 20+18+2.43+80$$
$$= 142.32\,\Omega$$

$\boxed{Y - \Delta}$



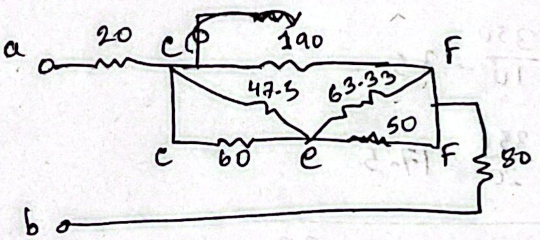
$$R_{cF} = \frac{30 \times 40 + 40 \times 10 + 10 \times 30}{10}$$

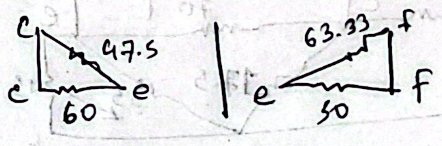
$$= 190$$

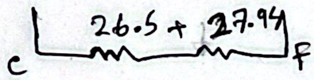
$$R_{eF} = \frac{1900}{30} = 63.33$$

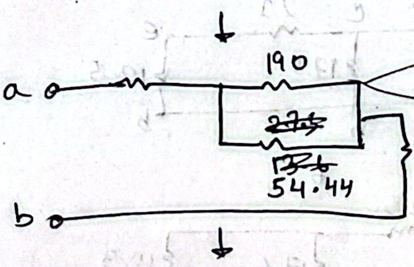
$$R_{ce} = \frac{1900}{40}$$

$$= 47.5$$

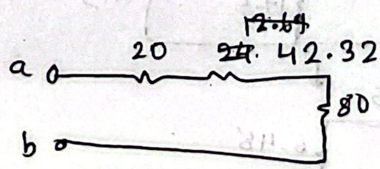






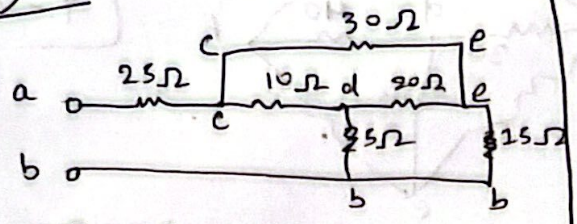


$$\frac{190 \times 54.44}{190+54.44}$$

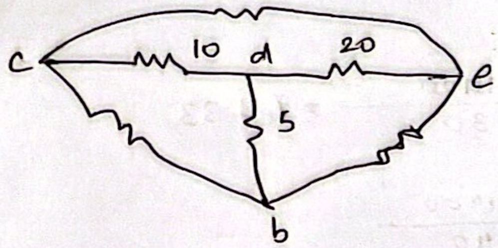


$$ER_{ab} = 80 + 42.32 + 20$$
$$= 142.32\,\Omega$$

2.51) b.



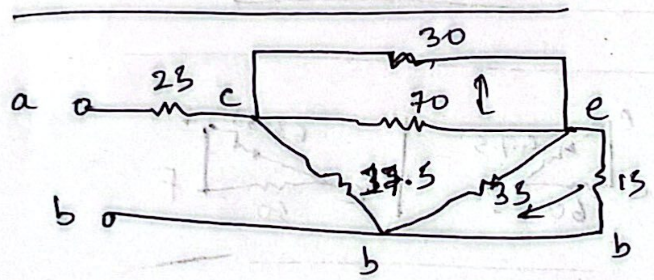
$ER_{ab} = ?$

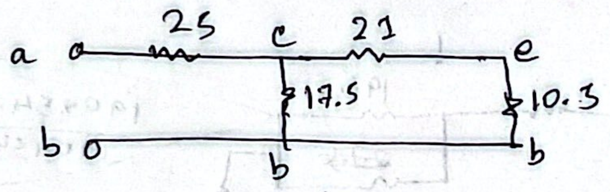


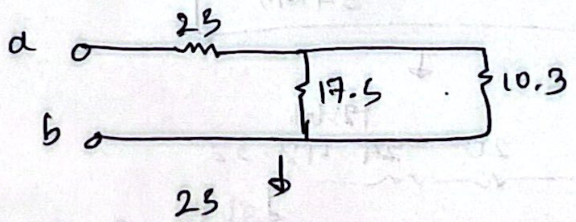
$$R_{ce} = \frac{10\times 20 + 20\times 5 + 5\times 10}{5} = \frac{350}{5} = 70$$

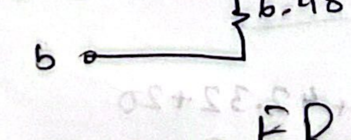
$$R_{eb} = \frac{350}{10} = 35$$

$$R_{cb} = \frac{350}{20} = 17.5$$









$$ER_{ab} = 25 + 6.48 = 31.48\ \Omega$$